



Information Technology Institute (ITI)

CMOS Analog IC Design

Self-Biased Sub-1V BGR Circuit

Design Challenge-1

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Part 1: BGR Core + Startup Circuit

Hand Analysis

Started by assuming that $V_2=V_1=0.7$ V

For current = $2\mu A$ in each branch Let

$$\Delta V_{BE} = \frac{kT}{q} \times \ln(n) = 81.88 \text{ mV (at Room temp)}$$

$$R_0 = \frac{\Delta V_{BE}}{1\mu} = 82 \text{ k}\Omega$$

$$R_1 = R_2 = \frac{V_{BE}}{1\mu} \text{ (Approximation current won't be the same in both branches)} = 700 \text{ k}\Omega$$

For PMOS Current mirrors Assumed High length = $4 \mu m$, $V_{DS} = 0.5$, $g_m/id = 10$ and $I_D = 2\mu A$

From ADT SA:

$$W = 10.56 \mu m, \quad L = 4 \mu m, \quad V_{GS} = 782.5 \text{ mV}$$

Startup Circuit Sizing

The startup circuit is designed to minimize power consumption after the OTA is operational.

- **M5 (PMOS):** Sized with minimum width and maximum length this ensures that only a very small current (picoamp range) is injected into the startup node consuming negligible power during normal operation.
- **M4 (NMOS):** Sized with a moderate width of $5\mu m$ and minimum length. This provides sufficient strength to pull the gate node of the bias current source to ground during startup ensuring the OTA does not remain in the zero current state.
- **M3 (NMOS):** Sized with $L = 1\mu m$ and $W = 5\mu m$. This matches the current requirements of the OTA bias.

The resistor values were optimized via DC sweeps to achieve the final set:

$$R_3 = 446 \text{ k}\Omega, R_2 = R_1 = 667 \text{ k}\Omega, \text{ and } R_0 = 103.5 \text{ k}\Omega.$$

Sizing Table

Table 1: PMOS Current mirrors and startup circuit transistor sizing

Device	Type	L (μm)	W (μm)
M_0, M_1, M_2	PMOS (CS)	4	25.92
M_5	PMOS (Startup)	5	0.45
M_3	NMOS (Startup)	1	5
M_4	NMOS (Startup)	0.6	10

Schematics with device sizing

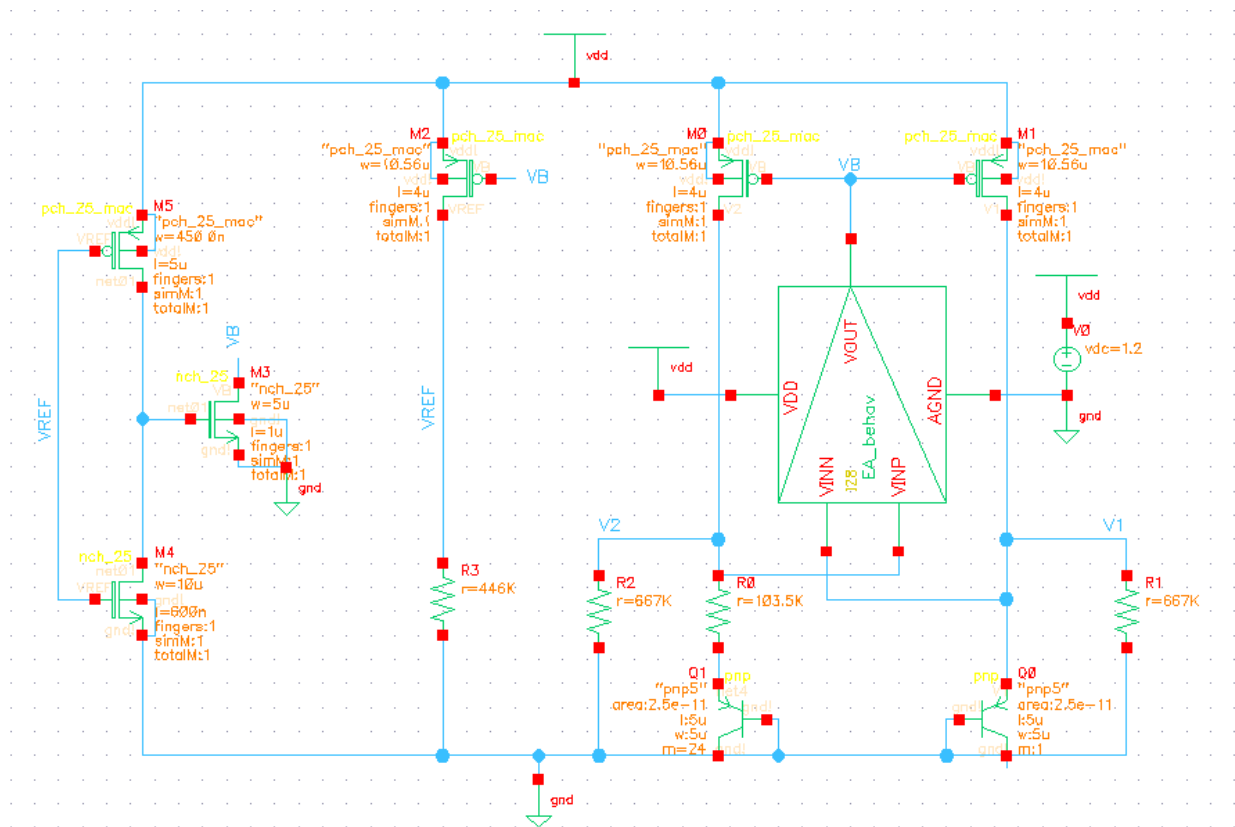


Figure 1 BGR TB Schematic

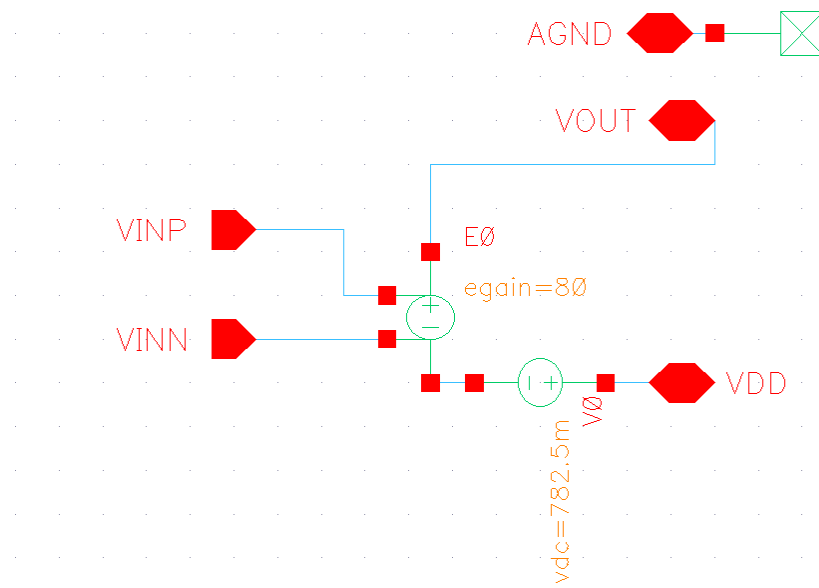


Figure 2 Behavioral EA Schematic

DC Operating Point

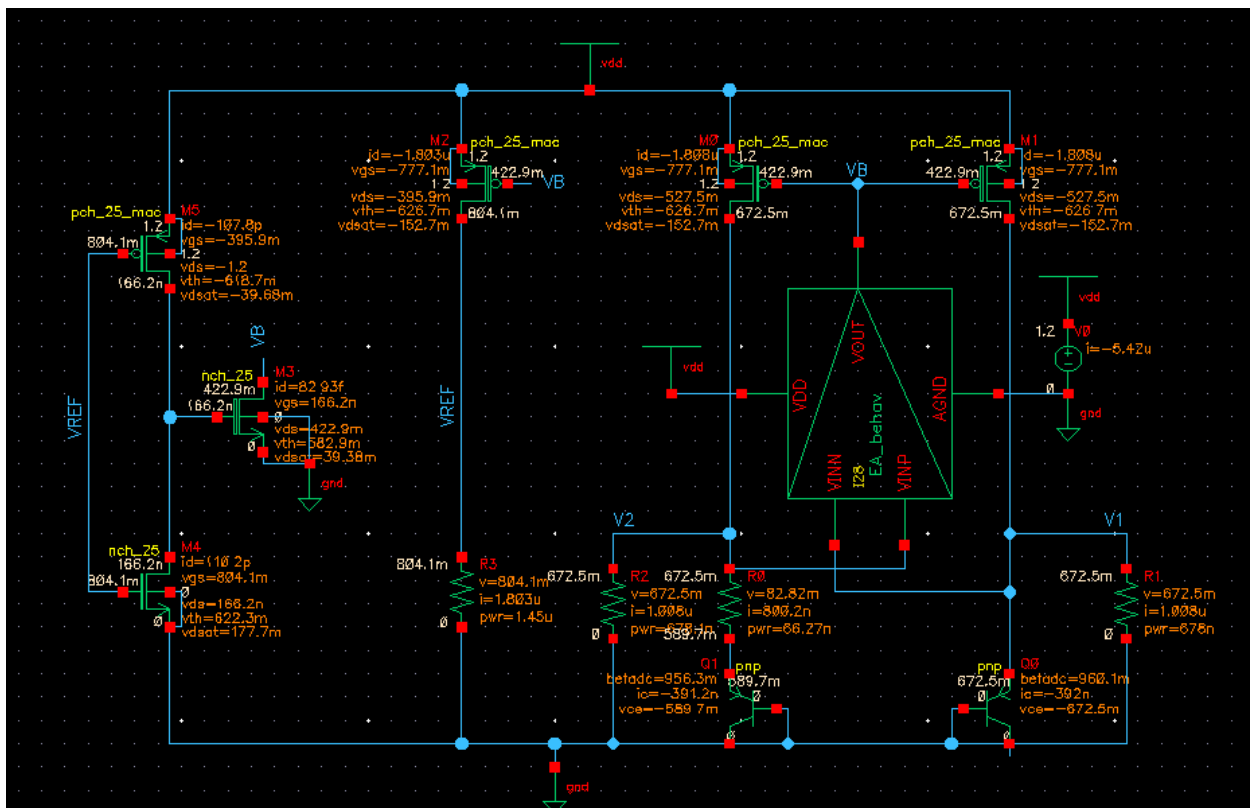


Figure 3 BGR core with DC node voltages and transistor OP parameters

Temperature Simulation Results

DC Analysis `dc`: temp = (125 C -> -40 C)

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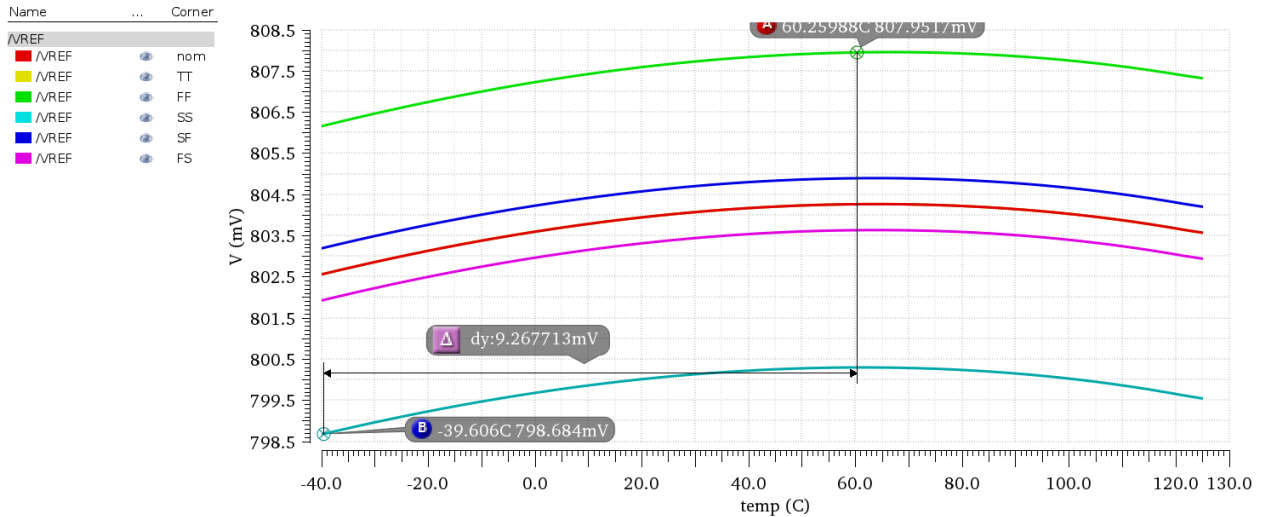


Figure 4 V_{REF} vs temperature at TT, SS, FF, SF, FS corners

Comment: The total mismatch is 9.26mV which is less than 10mV change to decrease this mismatch the OTA error amplifier design will be designed for a higher gain than 80.

Part 2: Actual Self-Biased Amplifier

Amplifier Design

Hand Analysis

- **PMOS current sources (M0/M1/M2)(BGR):** Sized with $L = 5 \mu\text{m}$. From ADT SA with $g_m/I_D = 10$, $I_D = 2 \mu\text{A}$, and $V_{DS} = 0.5 \text{ V}$: $W = 12.96 \times 2 = 25.92 \mu\text{m}$.
- **PMOS loads (M7(OTA), M2(OTA), M3(OTA)):** Sized as half of the PMOS current sources to match the OTA bias current and ensure V_{GS} matching with M7.
- **NMOS current sources (M5/M4)(OTA):** $I_D = 1 \mu\text{A}$ with $g_m/I_D = 12$, $L = 5 \mu\text{m}$, and $V_{DS} = 0.075 \text{ V}$: $W = 6.2 \mu\text{m}$ (from ADT SA). M4 has double the area to provide $1 \mu\text{A}$ to each OTA branch.
- **Input pair (M8/M9):** Sized with $L = 5 \mu\text{m}$ and $g_m/I_D = 18$ to achieve low V_{GS} . With $I_D = 1 \mu\text{A}$ and $V_{DS} = 0.5 \text{ V}$: $W = 11.52 \mu\text{m}$ (from ADT SA).

The amplifier gain is:

$$A_v = g_{m,8,9} \cdot (r_{o,8,9} \parallel r_{o,0,1,2})$$

High gain is achieved since the input pair provides high g_m and high r_o , while the PMOS loads are sized for high output resistance.

The self-biasing ensures correct DC operation without an external bias voltage.

Sizing Table

Table 2: Self-biased amplifier transistor sizing

Device	Type/Role	L (μm)	W (μm)	I_D (μA)
M0/M1/M2	PMOS (Current Source)	5	25.92	2
M7/M2(OTA)/M3(OTA)	PMOS (Load)	5	12.96	1
M5	NMOS (Mirror transistor)	5	6.2	1
M4	NMOS (Current Source)	5	12.4	2
M8/M9	NMOS (Input Pair)	5	11.52	1

Schematics with device sizing

The behavioral amplifier (VCVS) is replaced with an actual self-biased amplifier.

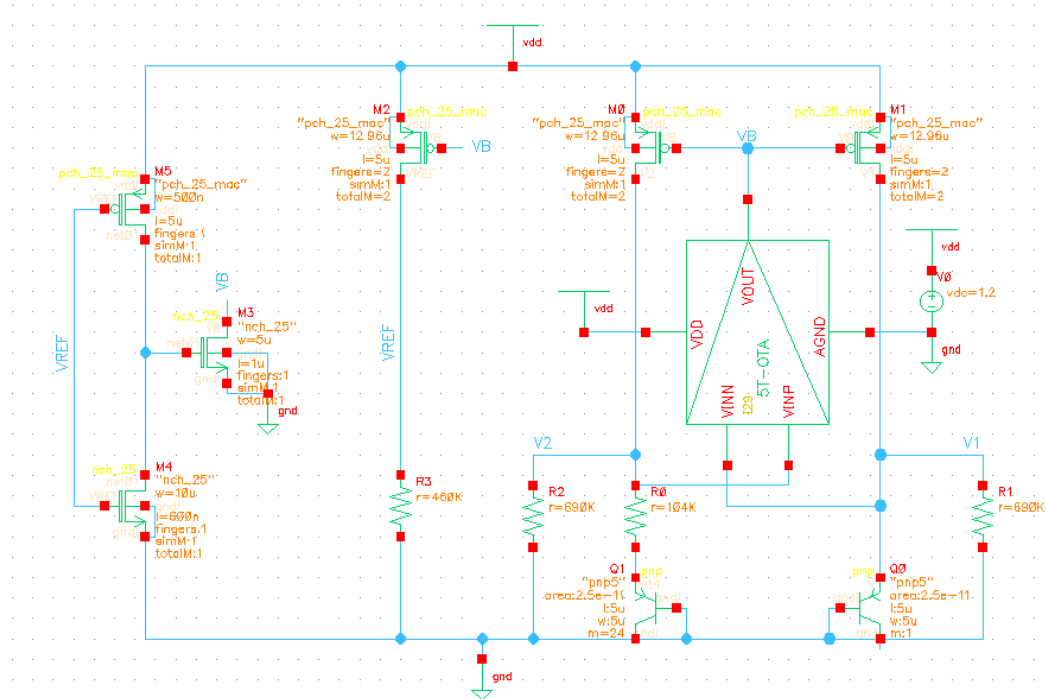


Figure 5 BGR TB Schematic

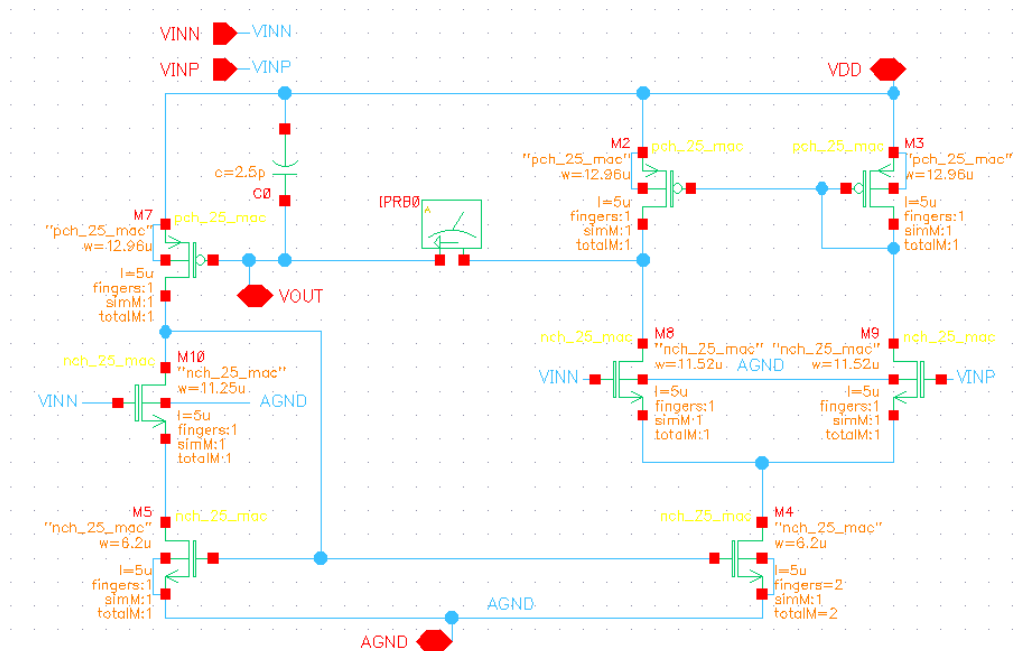


Figure 6 Error Amplifier Schematic

The diagram illustrates a complex multi-stage amplifier circuit. Key components and their parameters include:

- MOSFETs:** M1-M6 are NMOS transistors. Parameters like I_D , V_{GS} , V_{DS} , V_{th} , and V_{DSAT} are specified for each.
- Resistors:** R1-R4 are used for biasing and load. Values range from approximately 815.4 kΩ to 672.5 kΩ.
- Capacitors:** C1-C4 are coupling or bypass capacitors with values around 123 pF to 1 nF.
- OTA Core:** A central block labeled VDD, VOUT, AGND, VINN, and VINP, likely representing an Operational Transconductance Amplifier.
- Power Supply:** The circuit is powered by VREF, vdd, and connected to ground (gnd).
- Nodes:** Nodes V1-V8 represent various internal signal and bias points.

[illegible]

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Temperature Simulation Results

The complete BGR with the actual amplifier was simulated across temperature from -40°C to 125°C at all process corners.

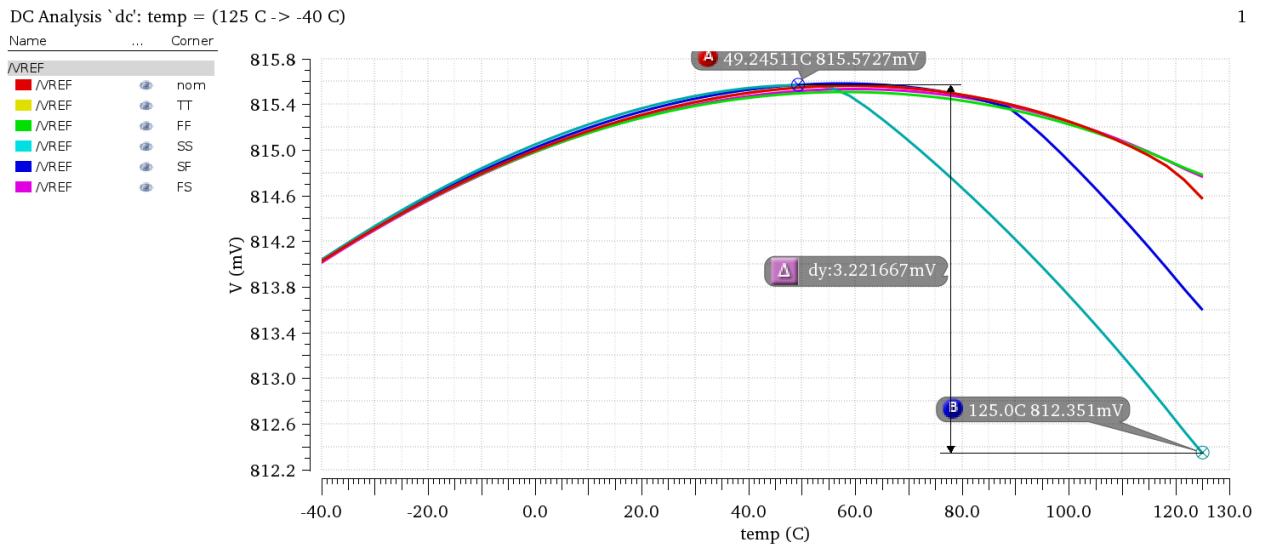


Figure 9 V_{REF} vs temperature (actual amplifier) at TT, SS, FF, SF, FS corners

STB Analysis

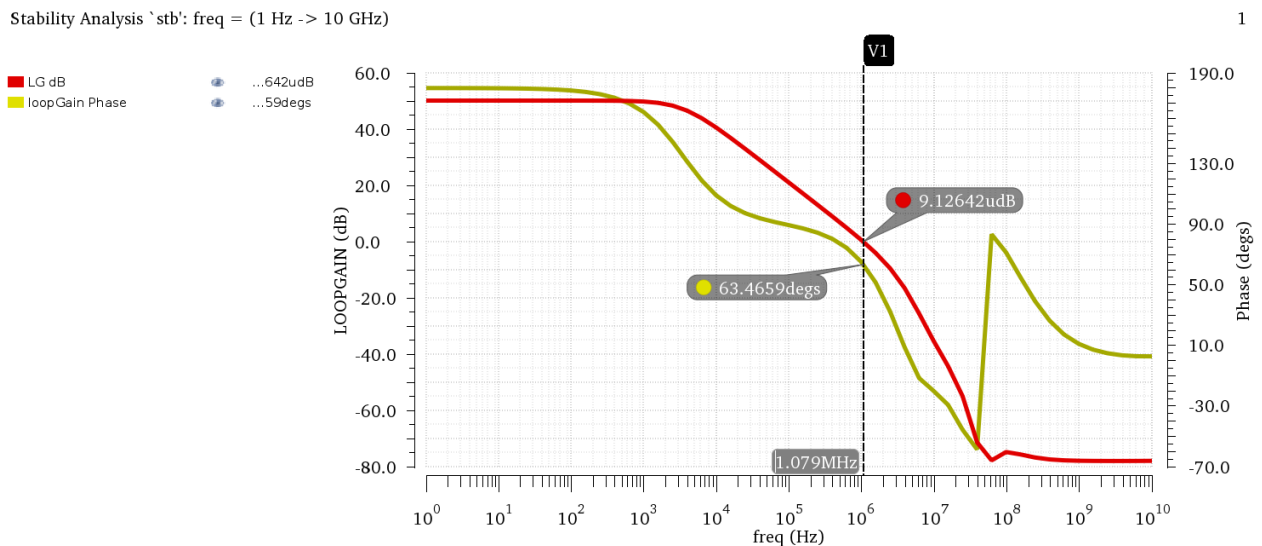


Figure 10 STB analysis: loop gain magnitude(dB) and phase vs frequency

Table 3: STB analysis results

Parameter	Value	Unit
DC Loop Gain	50	dB
Phase Margin	63.46	°

Comment:

The amplifier has a phase margin of 63.46° after compensation using 2.5pF capacitor.

Transient Simulation

The startup behavior was verified by applying a ramp to V_{DD} from 0 V to 1.8 V over 1 ms.

Transient Analysis `tran': time = (0 s -> 3 ms)

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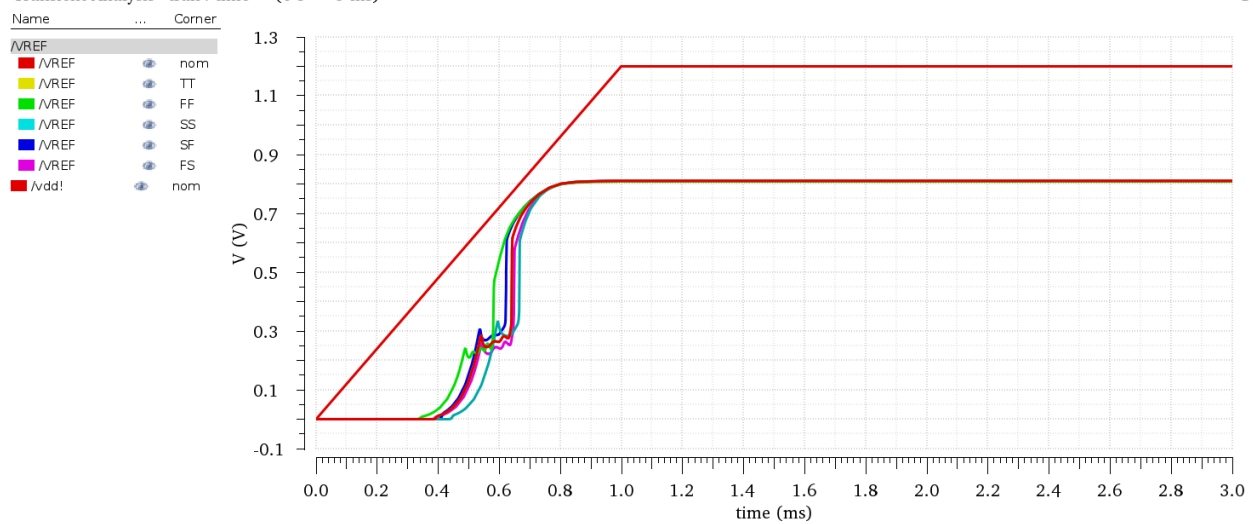


Figure 11 Transient simulation: V_{REF} vs time during V_{DD} ramp-up

Actual Passives

Passive Device Selection

The ideal resistors and capacitors were replaced with actual PDK passives. The following devices were used:

Table 4: Selected PDK passives

Device	Type	Value	Unit
R_0	Poly Resistor	463.6	k Ω
R_2, R_4	Poly Resistor	688.356	k Ω
R_5	Poly Resistor	99.865	k Ω
C_C	MIM Capacitor	2.5	pF

Schematics with device sizing

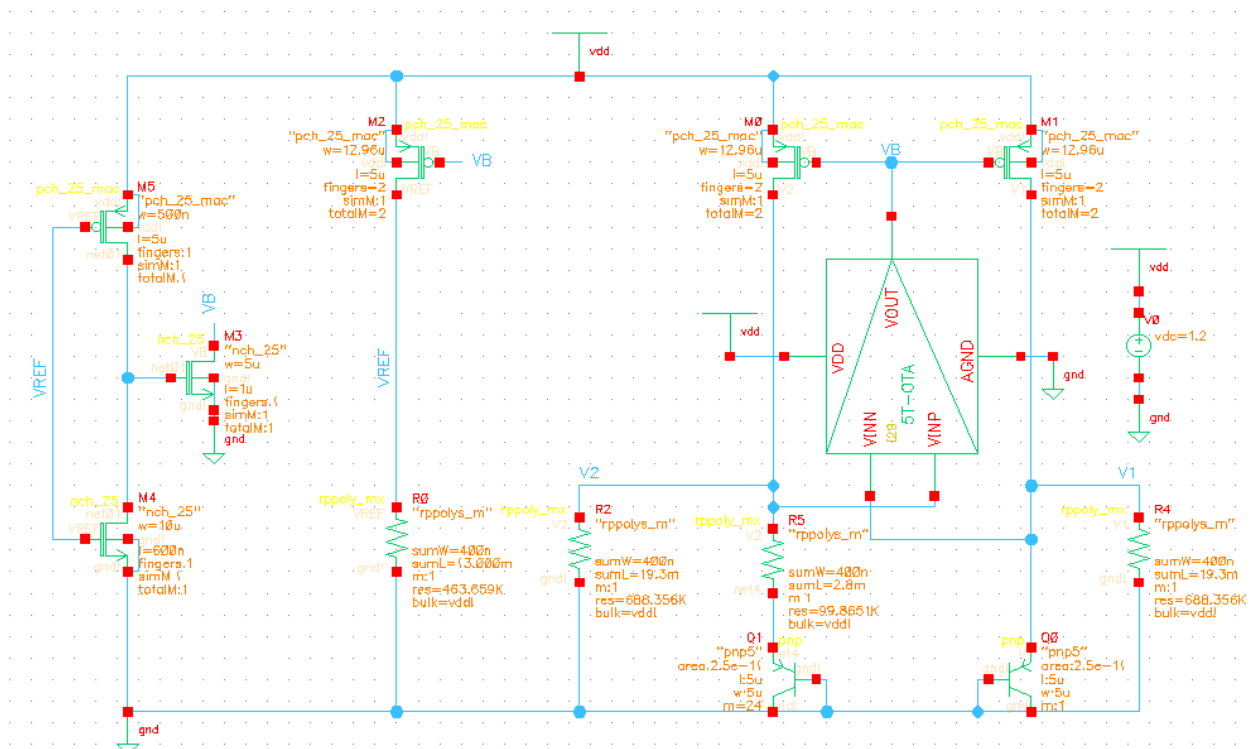


Figure 12 BGR TB Schematic

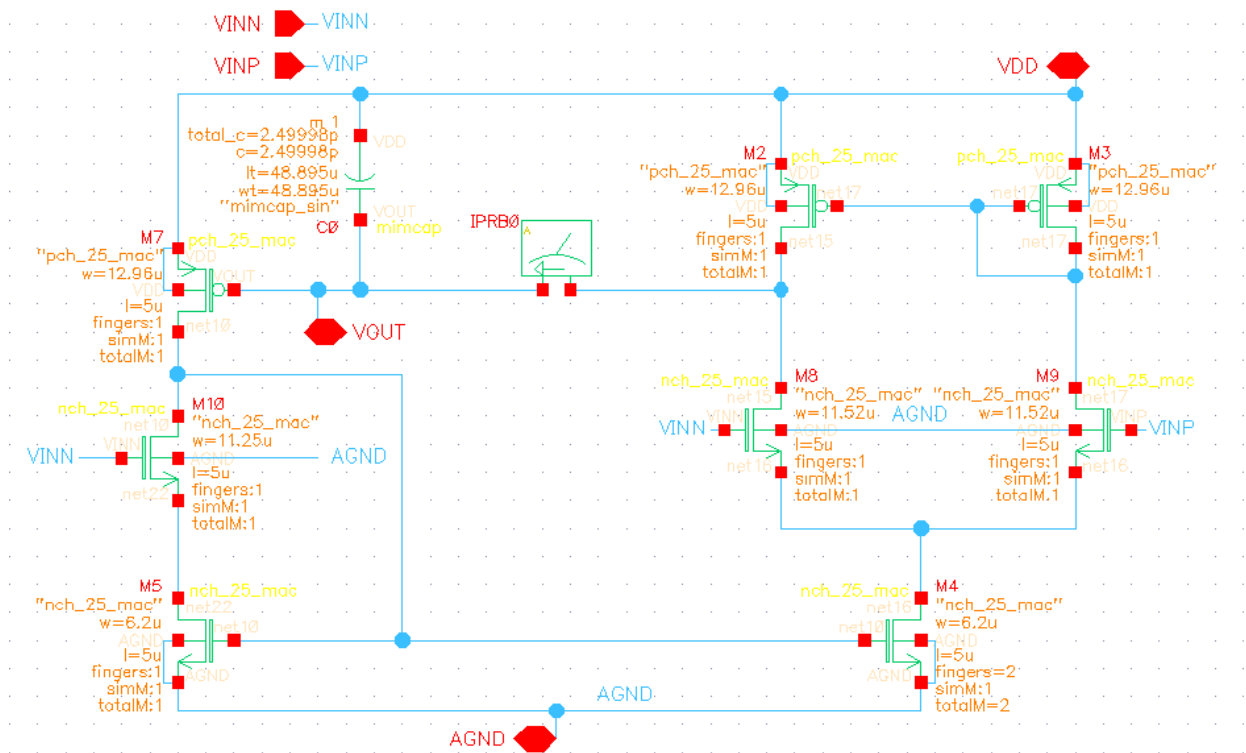


Figure 13 Error Amplifier Schematic

DC Operating Point

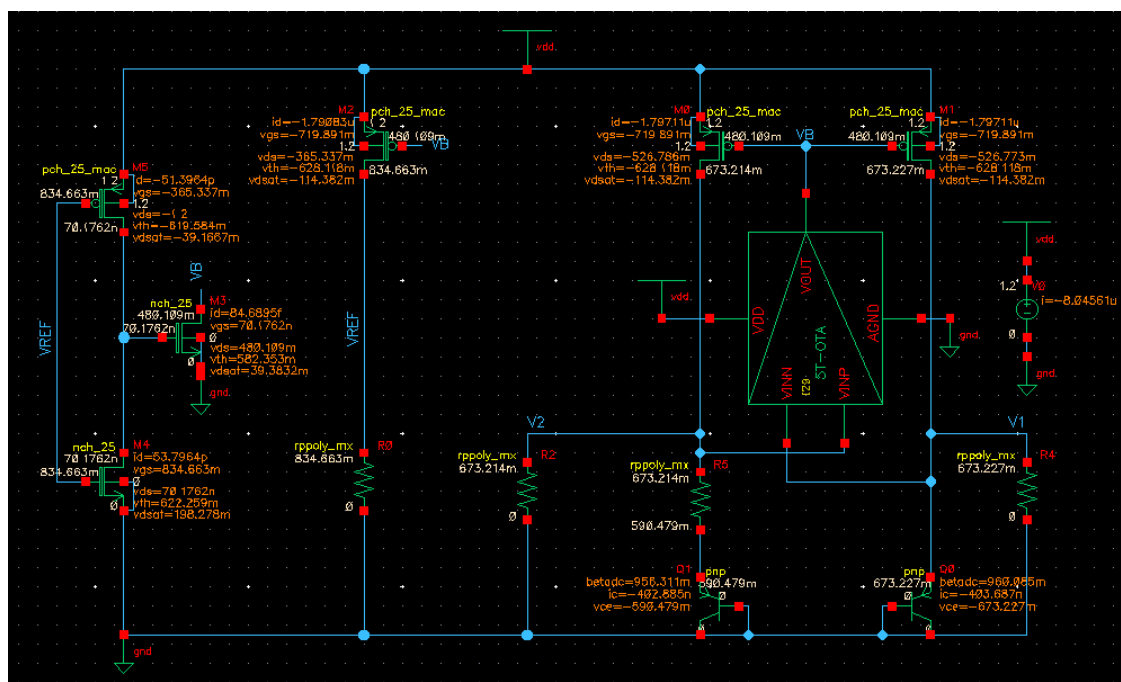


Figure 14 BGR with actual passives - DC node voltages

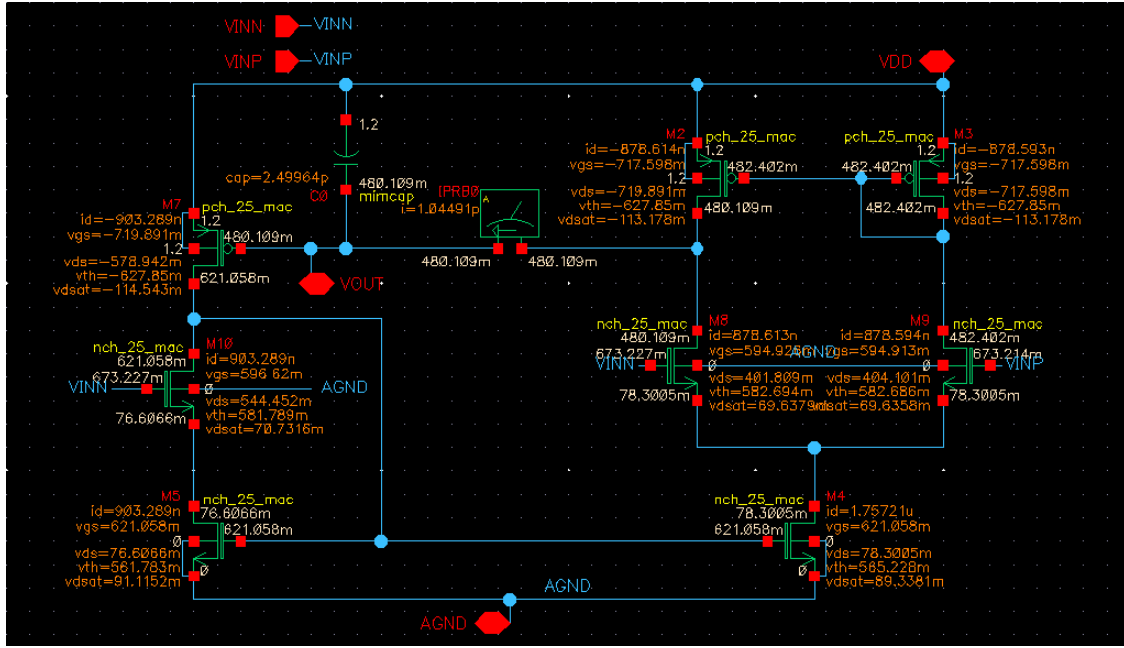


Figure 15 EA OTA with actual passives - DC node voltages

Temperature Simulation Results

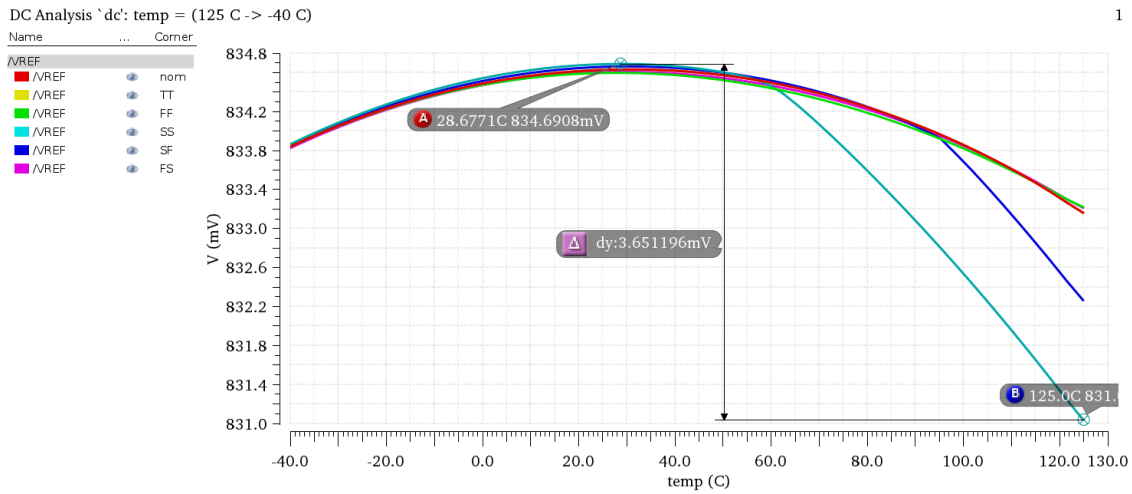
Figure 16 V_{REF} vs temperature (actual passives) at TT, SS, FF, SF, FS corners

Table 5: Design summary and specifications comparison

Parameter	Specification	Part 1 (VCVS)	Part 2 (Actual Amp)	Part 3 (Passives)
V_{REF} @ 27°C (TT)	0.8 V	0.803 V	0.814 V	0.83 V
Phase Margin	> 60°	> 60°	63.46°	> 60°
Total Bias Current	< 10 μ A	5.42 μ A	7.96 μ A	8.04 μ A
Amplifier Gain	-	80 V/V	316 V/V	316 V